

SIGN LANGUAGE RECOGNITION SYSTEM USING MEDIAPIPE AND LSTM

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ABSTRACT

This study develops a Filipino Sign Language (FSL) recognition system using Mediapipe hand landmark extraction and an LSTM classifier, achieving 94.5% accuracy on a 30-word FSL vocabulary.

INTRODUCTION

Deaf and hard-of-hearing individuals in the Philippines face communication barriers in educational and medical settings due to the shortage of FSL interpreters. Automated sign language recognition can bridge this gap. This study builds an FSL recognition system for the 30 most common conversation words, targeting real-time use on consumer laptop webcams.

METHODOLOGY

Mediapipe Hands was used to extract 21 3D hand landmark coordinates per frame (63 features). Sequences of 30 frames were collected for each of 30 FSL signs, with 60 sequences per sign (total 1,800 sequences from 10 signers). An LSTM network (2 layers, 64 units each) was trained on keypoint sequences with dropout=0.3. Evaluation: per-class F1-score, confusion matrix, and 5-fold cross-validation.

RESULTS

The LSTM achieved 94.5% accuracy on the test set. Per-class F1 ranged from 89% (THANK YOU, similar motion to SORRY) to 100% (simple static signs). Recognition latency averaged 0.11 seconds per sign. The system performed consistently across 8 of 10 signers; 2 left-handed signers showed 6% accuracy drop without mirroring augmentation.

DISCUSSION

Mediapipe's lightweight landmark extraction enables real-time FSL recognition on CPU-only devices. The left-handed signer gap was fixed post-evaluation by adding horizontal flip augmentation. Expanding the vocabulary to 100+ words is the primary next step, alongside integrating sentence-level context using a language model for more natural interpretation.